

CS 4610 HW 4

1. Given a random key K of length m bits. K is used as a one time-pad to encrypt plaintext message $P1$ (of length m bits) to obtain ciphertext $C1$. K is then also used to encrypt plaintext $P2$ (of length m bits) to obtain ciphertext $C2$ (i.e., the key is reused!). If the adversary is able to obtain $C1$, $C2$, and $P2$ (known plaintext attack), explain how he can compute $P1$. (10pts)

The adversary can easily recover the first plaintext message because reusing a key in a one-time pad negates its mathematical security. A one-time pad encrypts data using the XOR bitwise operation. Because the XOR operation is perfectly reversible, the adversary can take the known plaintext $P2$ and XOR it against its corresponding ciphertext $C2$. This operation cancels out the plaintext and reveals the original secret key K . Once the adversary has calculated the secret key, they simply perform the exact same XOR operation on the first ciphertext $C1$. By XORing $C1$ with the newly discovered key K , the encryption is reversed, and the adversary successfully recovers the unknown plaintext $P1$. Alternatively, the adversary could XOR both ciphertexts together to produce a combined string of $P1 \text{ XOR } P2$, and then XOR that result with the known $P2$ to isolate and reveal $P1$.

2. Alice wants to send some messages to Bob. Both Alice and Bob are aware that the communication channel between them is insecure. An adversary may modify the messages or impersonate Alice to Bob. They want to ensure the integrity of the messages. Answering the following questions.
 - a. How can they detect unauthorized modifications of messages only with secret key cryptography? Assume they have a way to share a secret key. (3pts)

To detect unauthorized modifications using only secret key cryptography, Alice can encrypt the entire message along with a predefined redundancy check or error detection code using their shared symmetric key. When Bob receives the ciphertext, he decrypts it using his identical copy of the shared key. If the decrypted message is readable and the redundancy check is mathematically valid, Bob can be confident that the message was not altered in transit. Because only Alice and Bob possess the shared secret key, an attacker cannot meaningfully modify the encrypted payload without turning the resulting plaintext into unrecognizable garbage upon decryption.

- b. How can they achieve this goal with hash function? Assume they have a way to share a secret key. (3pts)

To achieve message integrity using a hash function, Alice and Bob can utilize a Message Authentication Code. Alice will combine her plaintext message with their shared secret key and run this combined data through a cryptographic hash function. She then sends the original plaintext message along with the resulting fixed-length hash value to Bob over the insecure channel. Upon receiving the transmission, Bob takes the plaintext message, combines it with his own copy of the shared secret key, and runs the hash function himself. If his calculated hash perfectly matches the hash Alice sent, he knows the message was not tampered with. An attacker cannot alter the plaintext and generate a new valid hash because the attacker does not know the secret key required to compute it.

- c. How can they achieve this goal with public key cryptography. Assume they Bob knows the certificate of Alice. (4pts)

Alice and Bob can ensure message integrity and authenticity using public key cryptography by implementing digital signatures. Alice will first generate a hash of her plaintext message and then encrypt that hash using her own private key to create the signature. She sends both the plaintext message and the digital signature to Bob. Since Bob already possesses Alice's digital certificate, he can extract her verified public key and use it to decrypt the signature back into the

original hash. Bob then hashes the plaintext message himself and compares the two values. If the hashes match exactly, Bob has undeniable proof that the message was not modified and that it genuinely originated from Alice. This works because only Alice possesses the private key required to create a signature that her public key can successfully decrypt.

3. Programmer Joe is told to design a scheme to prevent messages from being modified by an attacker. Joe decides to append to each message a hash of that message. Why doesn't this solve the problem? (10pts)

Joe's proposed scheme fails to protect against malicious modifications because a standard cryptographic hash function is an entirely public algorithm that does not rely on a secret key. If an attacker intercepts the transmission, they can easily alter the plaintext message to whatever they want. Since the attacker knows exactly which hash algorithm is being used, they can simply compute a brand new hash for their forged message and replace the original hash attached by Joe. When the recipient receives the altered message and the forged hash, the two will match perfectly, tricking the recipient into believing the transmission is authentic. To truly prevent unauthorized modifications, Joe must incorporate a shared secret key into the hash computation to create a Message Authentication Code that an attacker without the key cannot forge.

4. What are the desired properties of hash functions? (10pts)

A secure cryptographic hash function must possess four primary mathematical properties. First, it must offer good performance, meaning it is computationally easy and fast to calculate the hash for any given message of any length. Second, it must have the one way property, meaning that if an attacker only has the resulting hash

value, it is computationally infeasible for them to reverse the mathematics to discover the original plaintext message. Third, it must be weak collision free, meaning that if an attacker possesses a specific message and its corresponding hash, they cannot find a different message that produces that exact same hash output. Finally, it must be strong collision free, meaning it is computationally infeasible for an attacker to find any two completely random and distinct messages that happen to produce the exact same hash output.

5. Hash functions are reasonably fast, but we can create a faster hash function. Take your message, divide it into 128-bit chunks, and then XOR the chunks together to get a 128-bit result. Apply the standard message digest (e.g., MD5 or SHA1) on the result. Is this a good hash function? Why or why not? (10pts)

This proposed method does not make a good cryptographic hash function because it violates the fundamental property of collision resistance. The XOR operation is mathematically commutative and highly predictable. If an attacker intercepts the message and wants to alter a specific chunk of data, they can easily do so without changing the final hash. To achieve this, the attacker simply flips the exact same bits in a different 128-bit chunk of the message. Because flipping a bit twice in a series of XOR operations mathematically cancels out the change, the final 128-bit XOR sum remains completely identical. Consequently, when this manipulated sum is passed to the standard message digest, it produces the exact same final hash value. Furthermore, an attacker could completely rearrange the order of the message chunks and the XOR sum would still not change. This makes it trivial for an adversary to forge or alter messages while maintaining a perfectly valid hash.

6. For a Diffie-Hellman key exchange, $p = 7$ and $g = 3$. If Alice sends to Bob the value 6 and Bob sends to Alice the value 5, what

is Alice's private key S_A , what is Bob's private key S_B , and what is the shared secret key on which they agree? (10pts)

For Alice, her transmission is 6, the generator is 3, and the prime is 7. By calculating the powers of 3 modulo 7, 3 to the power of 1 is 3, 3 to the power of 2 is 2, and 3 to the power of 3 is 27. Since 27 modulo 7 leaves a remainder of 6, Alice's private key must be 3. For Bob, his transmission is 5. Continuing the exponentiation, 3 to the power of 4 is 4 modulo 7, and 3 to the power of 5 is 243. Because 243 modulo 7 leaves a remainder of 5, Bob's private key must be 5. To determine the shared secret key, either party takes the other person's public transmission and raises it to the power of their own private key modulo the prime. Alice takes Bob's transmission of 5 and raises it to her private key of 3, resulting in 125. Calculating 125 modulo 7 yields a final remainder of 6, meaning the successfully negotiated shared secret key is 6.

7. RSA algorithms are widely used to generate public and private key pairs. Let e and d denote the public and private key respectively. Answer the following questions:

a. A public key is used to encrypt a plaintext M ($M < n$). What is the ciphertext? (10pts)

The sender raises the message to the power of the public key exponent e , modulo the shared modulus n . Because the message M must be mathematically smaller than the modulus n , this modular exponentiation scrambles the data into an unrecognizable format. Therefore, the resulting ciphertext is represented by the formula $C = M^e \bmod n$.

b. A private key is used to decrypt a ciphertext C . What is the decryption result? (10pts)

When a private key is used to decrypt a ciphertext C , the receiver raises the ciphertext to the power of the private key exponent d , modulo the shared modulus n . Due to the relationship between the public and private keys in RSA, this operation perfectly reverses the

encryption process. Therefore, the decryption result yields the original plaintext message, which is represented by the formula $M = C^d \bmod n$.

- c. A private key is used to sign a plaintext M ($M < n$). What is the signature? (10pts)

The sender raises the message to the power of their private key exponent d , modulo the shared modulus n . This does the same mathematical operation as decryption but applies it to the plaintext instead, creating a unique cryptographic receipt that only the private key holder could generate. Therefore, the resulting signature is represented by the formula $S = M^d \bmod n$.

- d. Alice sends a message M and the message's signature S to Bob. How can Bob verify that the received M and S are really generated by Alice? Assume that Bob knows Alice's public key. (10pts)

When Alice sends a message M and its signature S to Bob, Bob can verify her identity by performing the reverse mathematical operation using Alice's public key. Bob takes the signature S and raises it to the power of Alice's public exponent e , modulo n . This calculation will reverse the signature process and output a specific value. Bob then simply compares this resulting value to the plaintext message M that Alice sent. If the calculated value perfectly matches the message M , Bob can be absolutely certain that the signature was genuinely created using Alice's private key and that the message was not tampered with during transmission.